Procurement Technical Specification For Denver RTD EMU Communication System

Hyundai Rotem Company
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1. Introduction

1.1 General

This Procurement Technical Specification (PTS) defines the technical requirements for Communication system to be purchased by Hyundai Rotem Company and installed in the Electric Multiple Unit (EMU) Commuter Rail Cars, identified as the "Fastrack Eagle Project" cars, to be manufactured by Hyundai Rotem and to be operated by the DENVER Regional Transportation District("Owner"). The cars will be operated by DENVER RTD over the DENVER Metropolitan Area Railroad(East Corridor, Gold Line, Northwest Rail).

1.2 General Requirements

It is the intent that all Suppliers providing equipment or services to Hyundai Rotem shall comply with the Customer's requirements. When the Customer does not have any specific requirements, the Supplier shall comply with the requirements of this PTS as appropriate. Suppliers must comply with the requirements stated herein unless otherwise agreed to in writing by Hyundai Rotem. Any conflict between the Customer's requirements and this specification shall be brought to Hyundai Rotem's immediate attention for resolution. Notwithstanding conflict resolution, the most restrictive requirement shall apply. The term "Customer", as used in this document, refers to Hyundai Rotem's Customer (DENVER RTD).

This PTS in no way relieves the Supplier from any requirements specified in the Technical Specification.

All formal communication between Hyundai Rotem staff and Suppliers must be through the Procurement function at Hyundai Rotem. Formal transmittal from Suppliers of drawings, procedures, reports, and analyses referenced in this document must be sent to the responsible Hyundai Rotem Procurement Representative defined in the instruction letter transmitting this PTS. Formal approvals and documentation from Hyundai Rotem to Suppliers will be sent via the responsible Procurement Representative.

The order of precedence for Supplier technical requirements shall be as follows:

1) The Purchase Order between Hyundai Rotem and the Supplier

2) The Contract between Hyundai Rotem and the Customer

3) Hyundai Rotem's PTS

4) DENVER RTD Technical Specification Contract

As a guide, the sections of the aforementioned documents are considered to contain specific and relevant requirements have been referenced in this document. However, this in no way diminishes the Suppliers' responsibility for compliance with the Technical Specification. The complete requirements are those found within this PTS document, the Technical Specification, the Contract, addendums, and the documents referenced therein. It shall be the Supplier's responsibility to assure that current revisions of the subject documents are being utilized, and to assure that equipment, documentation, and services furnished pursuant to this PTS are in full compliance with all applicable documents.

1.3 Train Configuration

The configuration of on unit is as follows:

(a) Total number of train; 25 train sets.

(b) Each train(Unit) will consists of MA and MB.

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2. Definitions and Abbreviations

This section includes a list of all definitions and abbreviations applicable to PTS, GS and TS.

2.1 Definitions

The following definitions are applicable to the PTS, Technical Specification for the Purchase of DENVER RTD (TS).

"**Contractor**" means the persons or person appointed by the Employer to undertake the execution of the works for Hyundai-Rotem. In order to avoid misunderstanding the roles of the Contractor in GS and TS, the term “Contractor” shall be read as “Subcontractor” for those clauses referred to in the GS and TS by the PTS.

"**Engineer**" means the personnel appointed by DENVER RTD for the purpose of the DENVER RTD Electrical Multipule Unit.

“**Subcontractor**” means a supplier of complete Communication System to Hyundai-Rotem and Contractor specified in TS with regard to Communication system. Subcontractor shall carry out the works in accordance with TS with regard to Communication system.

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2.2 Abbreviations

The following abbreviations shall apply to the PTS and TS.

GS: General Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

TS: Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars.

PTS: Procurement Technical Specification for DENVER project issued by Hyundai-Rotem

RTD: Regional Transportation District

NA: Not applicable

EMI/EMC : Electro Magnetic Interference / Compatibility

S.A : System Assurance

RAMS : Reliability Availability Maintainability System safety

PDR : Pre-Design Review

FDR : Formal Design Process

3. Scope of supply and Work

3.1 Hardware

The Subcontractor shall be responsible for the provision of all equipments for the Communication System. The system shall comply the DENVER RTD specification (CONTRACT PART I. 3.17.1) and the following specification or more. Subcontractor shall provide all components related to Communication system, but not limited to, the followings:

1) Radio

1-1) Integrated radio control unit.

1-2) ~~Mobile~~ VHF radio transceiver.

1-3) Rooftop antenna.

1-4) Interconnecting wiring shall be Teflon high temperature jacketed type RG-303/U MIL-C-

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17D coaxial cable with approved connectors.

1-5) Radio power supply unit with suitable interference suppression line filter.

1-6) Suitable apparatus in the DENVER control center for the train dispatcher to receive and display this car number information.

~~1-7) Radio transmit control pushbutton, located on the right side of the cab center console outboard of brake controller. If possible, should be redundant at left hand.~~

1-8) Programmable Automatic-Time-Out (ATO) function

1-9) Audio recorder (Section12.1(a) of Attachment 7, Part C)

2) Public Address (PA) shall measure the local ambient noise level between use and automatically compensate gain accordingly and instantly just prior to every announcement.

2-1) Push-to talk button

2-2) Speaker selection Switch (3 positions)

2-3) Microphone

2-4) Power Amplifier

2-5) Speakers

3) Automatic Vehicle Location System (AVL)

3-1) Global Positioning Satellite System (GPS)

3-2) Wireless LAN (WLAN)

3-3) Dead Reckoning System for operation within the 1.5mile center City Commuter Connector tunnel plus other short highway overpass sections, which will provide the geographical location of the vehicle.

3-4) Vehicle Location Control Panel in each cab.

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4) Passenger Emergency Intercom (PEI), one at each of the two wheelchair locations and one at the doorway area at the end of the car opposite the wheelchair locations.

Each cab shall be equipped with two passenger emergency intercom system momentary contact pushbuttons or rocker switches, identified as “Passenger Intercom Push to Talk” and “Passenger Intercom Reset”.

5) Communicating Buzzer System

6-1) One at each Conductor’s doorway control panel, both upper and lower panels, for a total of 8.

6-2) One in each cab, for a total of either 1 or 2 per car.

6-3) One located at each end of the car, adjacent to the body end, for a total of 2.

6-4) One located on the car exterior at each low level doorway, easily reached when standing on a low level station platform, for a total of 4.

6-5) One located on the car exterior on each side of the end of the car, at each cab end of a car, easily reached when standing on the car during coupling/uncoupling, for a total of either 2 or 4 per car.

6) Cab Text Messages

7) Wireless Network

8) Wayside Installations

9) Communication Control Panel in each Cab, which has full function (control of PA, IC PEI and buzzer, and all indicator LEDs to signal activation)

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10) Conductor Communication Control Panel at Each doorway, which has PA, IC and buzzer function. This shall have separate momentary contact switches and indicator LEDs for control of the PA and IC functions. This panel shall also include a low profile, noise canceling microphone, and an IC speaker, both behind a perforated grille. A communication buzzer and push button shall also be included.

11) Key switch

12) Front Destination Indicator

13-1) Front, one exterior end destination

13-2) Side destination indicator shall be provided at each side doorway, for a total of 4 per car.

13-3) Interior, at least 6 display units. Two is at the front end of each section.

(* Note: The interior destination sign system shall be configured to permit the future connection to an active message system to be provided by DENVER for current and news and other information. And the subcontractor shall provide the interface schematic diagram for the active message system for carbody wiring).

13-4) Destination Sign Control Panel can be combined with Vehicle Location Control Panel

13) Accessory; All accessory for the communication system shall be provided by the subcontractor. The accessory includes pug connectors, pins or sockets, special cable for communication system, and fixing bolts & nuts for wayside equipment, and etc.

H/R Comments (16/11/2010)

1) Design shall be based on the altitude conditions and climatic conditions of the Denver area.
2) Equipment to be housed in equipment houses shall be rated for an operating temperature range of -40°F ~ 160°F at 0% to 95% relative humidity, non-condensing.

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3.2 Software

The Subcontractor shall provide, as a minimum, the followings:

1) Automatic Announcement

The Subcontractor shall provide all source codes, application software, mapping software and hardware required to make initial and future announcement changes, and shall disclose to all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses required to permit DENVER RTD to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost.

2) Programming Equipment for Radio

Five complete sets of radio programming equipment shall be supplied for setup and maintenance of the car radio transceiver by DENVER RTD. The subcontractor shall comply the detail requirement in the TS.

3) Destination Sign System

The subcontractor shall provide all source codes, software and hardware required to make initial and future sign system revisions, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses or other fees required to permit DENVER RTD and Hyundai-Rotem to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost to Hyundai-Rotem. All details of the destination signs system and operation of the control system shall be presented to Hyundai-Rotem and the Engineer of DENVER RTD.

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4) Automatic Vehicle Location System

The subcontractor shall prepare all mapping and survey work (location database) necessary to establish trigger points for all announcements, and prepare all software for complete operation. All mapping and survey work performed pursuant to this contract shall be the property of Hyundai-Rotem and DENVER RTD, and shall be delivered to Hyundai-Rotem in a format that is transportable and/or convertible in furtherance of DENVER RTD's interests. The subcontractor shall supply all required software and special equipment by a site license agreement as part of this Contract. The subcontractor shall provide all source codes, mapping software and hardware which is required to make initial and future announcement changes, shall prepare and install all updates and revisions to the cars for the basic warranty period for each car, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system. All required licenses required to permit to operate, maintain and modify the system for the lifetime of the car shall be included at no additional cost. There shall be no additional or reoccurring costs incurred by Hyundai-Rotem for any software, documentation, or special equipment required to operate, program, update, change displays and/or announcements and maintain the system as delivered. This shall include all software and hardware necessary to incorporate future changes and revisions in routes, destination displays, announcements, public relations messages and so forth.

The system in all operating modes shall indicate vehicle location to within 3 meters horizontally at 95% probability and the WAAS-enabled GPS system shall provide accuracy sufficient for track identification. The system and components shall be designed to permit integration with current digital and analog-type radio systems and cell phone networks. The subcontractor shall provide all source codes, mapping software and hardware along with a complete bill of material. The subcontractor shall disclose to DENVER RTD all required proprietary methods, software and tools and associated equipment necessary to ensure the successful operation of the system. All aspects of the system shall be in compliance with the Americans with Disabilities Act as amended.

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5) Expansion Provisions

An Automatic Vehicle Location (AVL) and announcement system utilizing an open architecture shall be supplied to provide for future expansions and upgrades to the system. The subcontractor shall provide all source codes, mapping software and hardware required to make initial and future changes, and shall disclose all required proprietary methods, software, tools and associated equipment necessary to ensure the successful operation of the system.

6) Wireless Local Area Network

All WLAN systems, the Denver RTD Control Center and selected management department shall be connected via a secure extranet. Associated servers and networking software shall be supplied by the subcontractor. Extranet fees be borne by the subcontractor for the full term of the contract (last car of warranty).

7) Non-commercially available software

The subcontractor shall provide the all non-commercially available software defined in the TS.

8) Commercially Available software

All commercially available software shall be listed-up by the subcontractor.

3.3 Wayside Installation

The Subcontractor shall provide all hardware and engineering work for the wayside equipment, as a minimum, the following:

1) Radio

The subcontractor shall provide, install, test and verify all of the equipment in the DENVER RTD Control Center required to receive, decode and display the data transmitted by the vehicle radio

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system required under DENVER RTD technical specification. The subcontractor shall install in the Control Center all equipment required to allow the ten train dispatcher consoles to communicate to train passengers via the public address system on the cars. This will allow the train dispatcher to give public service announcements directly to the passengers. This shall be done by selecting a particular car or cars, all cars or all cars on a selected route, using apparatus approved by the Engineer. There shall be no more than a 2 second delay for any variation of car selection prior to the system permitting the announcement to be made. Details shall be submitted to the Engineer for approval.

2) CAB Text Messages

The subcontractor shall design and supply a complete system to permit electronic transmission of written Form movement permits from the DENVER RTD EMU dispatcher to the Operator in the cab during all times when the train is in operation. This shall include the selection of proper radio transmission methods, the installation of all equipment in the DENVER RTD Control Center and all other wayside apparatus as required, all carborne equipment, and all licenses, permits and other registration fees required for a period of at least 20 years from the delivery of the last car.

3) Wireless Local Area Network

As part of this Contract of Denver RTD project, the subcontractor also supply and install wireless local area networks (WLAN) in DENVER RTD project, (including all hardware, software, extranet and monitoring location equipment as follows):

Individual WLAN locations shall be installed at each of DENVER RTD's Railroad Division five (5) yards and shops along with one (1) location logistically central to train operations such as DENVER RTD's Suburban Station, for the primary purpose of automatic fault data exchange between DENVER RTD center and the DENVER RTD EMU along with automatic file update downloads for the AVL system. The facility based WLANs shall also accommodate other maintenance management and information software used by DENVER RTD so they are accessible via the WLAN networks.

All WLAN location data systems shall be configured to automatically capture the car number, fault and status information from each Denver RTD EMU as it enters within any WLAN area. All fleet status information shall be integrated into a real-time information software system with monitoring capabilities at each yard, the DENVER RTD Control Center and one key management location through a common extranet provided by the Contractor.

The subcontractor shall design the topography of the network as applicable for each location and supply and install all needed hardware and software. The IEEE802.11 technology chosen for the WLAN shall be based on the site surveys for network coverage concerning reception such as signal-to-noise ratios, EMI interferences, speed potential and estimated user bandwidth loads. An adequate number of wireless access points and associated antenna shall be provided to accommodate efficient speed and access based on the use of the proposal automated data transfer system, DENVER RTD's Vehicle Maintenance Information System (VMIS) and Vehicle Technical Information Library (VTIL) system.

3.4 Spares, Special Tools and Testing Equipment

The Contractor shall furnish the following capital spares, test equipment, and special tools. If a Rolling Stock Option is exercised, the proportional number of capital spares will be provided. Selected special tools and test equipment shall also be required depending on the nature of the tools/equipment and its intended usage.

3.4.1 Special Tools and Testing Equipment

All tools, excluding tools commonly available from standard suppliers such as Granger and/or Snap-On that are required or desirable for the effective/efficient conduct of maintenance and

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repair of the vehicles will be provided as a "special tool" in quantities necessary to support the level and intensity of vehicle maintenance anticipated. This requirement extends through mechanical, electrical, electromechanical, and electronic tools, including software products. These tools will be provided within 60 days of the delivery of the first production Car of the Base Order.

All test equipment required or desirable for the conduct of maintenance and repair activities of the Car will be provided in sufficient quantity to adequately support the maintenance operation. All such test equipment to be provided by the Contractor regardless of the equipment's availability from other common sources. Portable test equipment (Two of each kind, including instruction manuals, cables, connectors and associated software), shall be provided for all on-board systems. All portable test equipment shall be delivered within 30 days of the delivery of the first production Car of the Base Order. One set of bench test equipment shall be provided for the purpose of testing, troubleshooting and calibrating all electrical, electronic, mechanical and electro-mechanical components of each vehicle subsystem. All test equipment, including bench test equipment, shall be delivered within 90 days of the delivery of the first production Car of the Base Order.

The Subcontractor shall provide, at a minimum, the following:

The spare parts shall be based on the principal that local maintenance will be to Level One. This level assumes the diagnosis and replacement of major parts and line replaceable units (LRU) locally. Defective units and/or their sub-assemblies will be returned to the subcontractor for repair.

The Subcontractor shall recommend and submit the detailed list of spare parts for maintenance.

The Subcontractor may recommend additional spare parts and separately quote spare parts if he esteems necessary for maintenance and repair of the supplied Lighting for Hyundai Rotem's further consideration as separate purchase.

The Subcontractor shall propose the spare parts according to Section 20.12 Spare Parts and provide the following parts as a minimum, according to proposed in BAFO submission.

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3.4.2 Spare

A minimum of 2 Car sets of all replaceable/rebuildable interior body units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to all interior body panels and fixtures, seats, window assemblies, windshields assemblies door leafs, end doors, and other body parts as may be damaged and require replacement.

A minimum of 3 Car sets of all replaceable/rebuildable mechanical/electrical units will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. This provision will include but not be limited to fully dressed (built-up, ready to install) trucks, traction motors, air compressors, air conditioning compressors, air conditioning condensers, complete air conditioning units (including frame), propulsion equipment, auxiliary power equipment, train control equipment, communications/radio equipment, safety equipment, master controller, operators "desk" equipment, door operators and other major mechanical, electrical, and electromechanical equipment.

A minimum of 4 Car sets of couplers and draw bar assemblies will be provided as capital spares deliverable to the Site before the delivery of the last production Car of the Base Order. In addition, 4 compromise couplers will be provided.

3.5 Interface Responsibilities

The subcontractor shall provide recommendations and specifications on the installation of all the subcontractor's supply equipment. The subcontractor shall provide the interface specification between Communication system and vehicle equipments (as TS, but not limited to) and performed the interface test between Communication system and vehicle equipments. Hyundai Rotem will perform the

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installing design and installing work of Communication system with the subcontractor's information. The subcontractor shall develop and provide a performance criteria which shall include such items as maximum allowable percentage of false reports and maximum allowable percentage of missed failure. The subcontractor shall provide the weight and the power consumption of Communication systems.

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3.6 Design Information

The Subcontractor shall provide all necessary documents and deliverables for Hyundai-Rotem according to the time schedule defined by Hyundai-Rotem. The Subcontractor shall provide the interface information, which possibly affects performance, fitting, and form, from Hyundai-Rotem. The documents and the time schedule are listed and specified in part II of this specification "Design Information".

The Subcontractor shall provide the technical requirement and design information. (refer to section 4 and 5 of this PTS)

All drawing shall be provided by Auto-cad file.

3.7 Testing and Inspection

The Subcontractor shall be responsible for performance tests of the Communication System. The subcontractor shall be responsibility for a compressive series to tests to be performed to verify the suitability of design and workmanship of each car according to the requirement in DENVER RTD TS requirement (CONTRACT PART I, SECTION 4). And the subcontractor shall provide the test procedure and the test report after perform the each test. Hyundai Rotem and/or DENVER RTD have the right to witness any of these tests at the any stage of test progress.

Hyundai Rotem and/or DENVER RTD will carry out the First Article Inspection (FAI) on complete Communication System under the Subcontractor's responsibility. The Subcontractor shall carry out the type test and routine test of equipment and assembly. The commissioning test of the complete vehicle will be performed by Hyundai Rotem under the Subcontractor's responsibility for the Communication SYSTEM and coupling assembly. The Subcontractor shall provide the required information for testing and carry out the tests and retrofit problems during the tests in accordance with PTS.

In the event that any test for the Communication SYSTEM is failed, the Subcontractor shall, at his own expense, take whatever action is deemed such as, rectification, readjustment or design changes to the satisfaction of Rotem and DENVER RTD, in order to meet the testing requirements.

3.8 Operation & Maintenance Manual

The Subcontractor shall provide necessary requirements for Operation and Maintenance Manual and Spare Parts Catalogues for the Communication system and special tool with hard copies and electronic format.

The requirement for Operating and Maintenance Manuals and Spare Parts Catalogues shall be provided for Approval by Hyundai Rotem according to the time schedule approved by Hyundai Rotem. The Subcontractor shall provide softcopy for all O&M Manual according to detail requirement (Attachment 10) and required formats provided from Hyundai Rotem.

H/R Comments (16/11/2010)

1) Woojin shall provide the operation and maintenance manuals as per the TP23 and discussed information with Mr. John Henning at 16/11/2010.

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Denver RTD
Attachment 10 .pd

3.9 Training

The Subcontractor shall provide comprehensive training to the DENVER RTD Employer's staff (maintenance, operating, training instructor and engineering) in accordance with DENVER RTD TS. The training activities and works for the communication system shall be approved by Hyundai Rotem. The subcontractor shall prepare the training manual and provide it. The training manual shall follow the sample in the Appendix A "SAMPLES" of TS

H/R Comments (16/11/2010)

1) Woojin shall provide training (document & information) as per the TP23 and discussed information with Mr. John Henning at 16/11/2010.

3.10 Warranty

The subcontractor has sole responsibility under this Contract to maintain sufficient warranty spare parts in this stock to support the warranty period requirements, according to the DENVER RTD CONTRACT PART I, SECTION 5 or supply of equipment manufactured specifically for DENVER RTD.

4. Technical Specification

The Subcontractor, who will supply the communication system for Denver project, shall comply all sections related with Communication system of Technical Specification for the purchase of Electrical Multiple Unit Commuter Railcars (TS).

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4.1 Technical Requirements

The communication system can meet all technical requirement required in DENVER Technical Specification (TS).

The subcontractor shall review the below section of TS, as a minimum, and comply;

4.2 Electromagnetic Compatibility

The subcontractor shall ensure that the subsystem and components operate correctly and reliably within the Line operating electromagnetic environment, and shall not themselves materially and adversely affect adjacent third party infrastructure and systems. The subsystem shall be electromagnetically compatible with DENVER/Amtrak's signaling systems, with DENVER/Amtrak's communications systems and with adjacent installations.

EMC Control

The Contractor shall submit electromagnetic compatibility control and test plan for approval and the plan shall insure that proper emphasis will be placed on the control of interference and adequate attention given to interface and Federal Communications Commission (FCC).

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The subsystem shall operate without either suffering or causing interferences which may impact the system operation and/or safety because of conducted and/or radiated emissions and the contractor shall employ design techniques, construction methods, and whatever equipment is required to prevent interference caused by internal sources from affecting the proper operation of the car subsystem. In addition to coordinating frequencies, the Contractor shall provide necessary on-board balancing, filtering, shielding, modulating techniques and isolation to maintain signal to noise (S/N) ratios above limits. Shielding, isolating, balancing and grounding shall be used as required to reduce the undesirable effect of interference. Electrostatic and magnetic shielding methods shall be employed to minimize the effect of stray signals and transient voltages on interconnecting cables. Interconnecting power and signal cables shall be physically separated.

Cellular (wireless) telephones, typical farecards, credit/debit cards, laptop computers, PDAs and other magnetic storage media such as floppy disks shall not be adversely affected by EMI fields at any location inside and outside the vehicle, including the floor, ceiling, Operator's cab, or in proximity of equipment cabinets. For farecards, an exposure limit of 500 Gauss shall be considered as the standard for farecard susceptibility.

The Contractor shall ensure that any static or alternating fields generated by the subsystem shall not interfere with the correct operation of cardiac pacemakers, customer goods or any magnetic media. At a minimum the magnetic flux density in any car due to the subsystem shall not exceed the ICNIRP guideline elsewhere under any mode of operation, including failure modes.

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Emission Limitations

The Contractor shall ensure that the cars do not produce any radiated emissions that will violate FCC Rules and Regulations, 47 CFR Part 15.103. The radiated emissions of the Pilot Car shall not exceed the level permitted by the above standard at any frequency between 0.15 MHz and 1000MHz. Testing shall be based on the document "Radiated Interference in Rapid Transit Systems, Volume 2 - Suggested Test Procedures, UMTA-MA-06-0153-85-11" for guidance and shall be conducted to the satisfaction of the Engineer.

Safety Analysis

The Contractor shall prepare an EMI safety analysis which demonstrates the protection against hazards due to EMI. A draft analysis shall be submitted for review at the early Preliminary Design Review; an updated version shall be submitted at the mid-point in the design process between the Preliminary Design Review and the Final Design Review and the completed analysis shall be submitted at the early Final Design Review.

The Contractor shall document through the EMI safety analysis that the subsystem's EMI emissions are safe under normal conditions, and that the subsystem provides adequate detection, annunciation, and response to failures which could cause EMI to increase above acceptable levels. The EMI safety analysis shall include a Preliminary Hazard Analysis; a Fault Tree; and a Failure Mode Effects Analysis.

The EMI safety analysis shall document the protective actions which prevent a hazard occurrence. The EMI safety analysis shall demonstrate that the equipment and its actions are adequate to prevent an EMI hazard. The EMI safety analysis shall distinguish between failures which affect EMI and for

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which there is automatic protection, those which are annunciated for Operator or Maintainer action, and those which are undetected failures.

The EMI safety analysis shall consider all relevant failures of the subsystem, including failures of line filter components, EMI detection and annunciation systems, and any other components that can affect the level or frequency of the generated EMI; unusual operating conditions; and failures of wayside equipment affecting, such as a substation component failure. The EMI safety analysis must also include equipment self-testing in areas that affect EMI.

Compatibility Design Report

The Contractor shall prepare an Electromagnetic Compatibility (EMC) Design Report prior to Final Design Review, and subsequently update it when and if design changes occur as a result of testing. The report shall consist of two parts, Emissions Report and Susceptibility Report, as described below.

The Emissions Report shall document:

a) EMC emission control methods, including equipment layout, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding. The Contractor shall comply with IEEE Standard 1100-1992, Recommended Practice for Powering and Grounding Sensitive Electronic Equipment, unless otherwise approved;

b) Description of each unit with a peak power rating of greater than 5 kW, including the emission characteristics versus operating mode, voltage and loading, as well as line filter, output filter, line input impedance, and circuit and operating mode considerations;

c) Characteristics of any repeating narrow band emissions and frequency stability control measures.

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The Susceptibility Report shall document:

a) EMC susceptibility control methods, including equipment layout, equipment enclosure design, circuit routing, frequency coordination and stability, filters, interconnections, grounding, and shielding;

b) The high voltage and low voltage grounding arrangement, and provisions for worst-case ground or return bus voltages within the car;

d) The normal and fault paths for ground currents;

e) The equipment protection as well as any restrictions on equipment operation or maintenance necessary to ensure susceptibility control, such as not leaving covers off of equipment enclosures; and

f) For each type of electrical interface between subsystems or enclosures:

i) The driving circuit, including impedance and maximum and minimum output voltage and current levels;

ii) The receiver circuit, including maximum and minimum voltage or current thresholds, margin against error, and destructive threshold;

iii) All aspects of the connection requirements, such as single wire, double wire, twisted pair, shield, and shield grounding; and

iv) Any special circuit or connection provisions needed to ensure EMC.

Transient Protection

The subsystem shall be self protected from damage and improper operation due to high voltage transients across the supply terminals of that equipment, high voltages impressed between either supply terminal and the car body, including continuous imposition of the auxiliary supply level, and long term over voltage and under voltage conditions resulting from equipment failure modes.

Equipment powered by the low voltage power supply shall be capable of both withstanding a voltage

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increase of 30 percent above the maximum normal voltage for 0.5 seconds, and be capable of withstanding transients of 3.0 kV peak pulse with a total energy of 90 joules, without damage, improper operation or shutdown.

The voltage or current levels of electrical signals shall be sufficiently high to allow transient filtering without information loss. Adequate voltage transient suppression shall be provided for the protection of panels and circuits incorporating semiconductor devices.

Suppressors shall be incorporated across inductive devices to minimize switching transients. All magnet valves and relay/contactor coils shall have free-wheeling diode or metal-oxide varistor arc suppression. Other means of suppression or the absence of suppression for performance reasons shall have the approval of the Engineer prior to use. The number of types of suppression devices shall be kept at an absolute minimum. Equipment enclosures shall shield its equipment from any effects resulting from the operation of a 5 watt handheld high band VHF radio transceiver when used anywhere within 1 foot 6 inches of the enclosure.

4.3 Fire

The Fire Safety requirements shall be met, but not be limited to, the following the TS. Smoke and Flammability in Technical Specification:

(a) All applicable materials used in the manufacture of the cars shall meet the requirements of the Federal Railroad Administration, United States Department of Transportation as contained in 49 CFR Part 238.103 and its Appendix B.

(b) Third party laboratory certification must be presented to the Hyundai-Rotem and DENVER for all applicable materials. The supplier shall also have performed, for review and approval by the Hyundai-Rotem and DENVER, a fire safety analysis in accordance with 49 CFR Part 238.103 (c). Any exceptions shall be decided by the Hyundai-Rotem and DENVER. Conformance with the flammability and smoke emission standards is solely the responsibility of the supplier.

(c) All car components shall be designed for the prevention of fire, shall provide protection of the public, employees, and emergency response personnel from injury due to fire, smoke, explosion, or panic due to fire, and shall provide protection of system elements from damage by fire or explosion. Equipment that is a potential fire hazard shall be appropriately isolated from the passenger compartment. Enclosures for control and other critical equipment shall be located to provide protection against environmental contamination and mechanical damage.

(d) The use of polyvinyl chloride (PVC) or polyurethane plastic in the car, except as may be identified in this Technical Specification, is prohibited unless approved by the Hyundai-Rotem and DENVER.

(e) Wire Insulation

- Underwriters Laboratories Specification 1581, Section 1080, VW-1 Flame Test.

- Association of American Railroads Manual of Standards and Recommended Practices, Recommended Practice RP-585, Wiring and Cable Specification, plus the following special requirements:

a) Flammability

Irradiated cross-linked polyolefin shall meet the requirements of test method ICEA S-66-524, Paragraph 6.12.5, except that after five flame applications of 15 seconds each with a three-second rest period between applications, the flame shall extinguish in ten seconds or less.

b) Smoke Density

National Institute of Standards and Technology smoke chamber test using a 5.0 feet long piece of number 12 AWG 2000 volt wire. Neither the maximum specific optical density (Dm) nor the specific optical density after 4 minutes (Ds-4 min.) shall exceed 200 on a flaming test and 200 on

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a non-flaming test

(f) Safety

Passengers and crew shall not be exposed to tripping hazards, any electrical voltage, toxic materials, abrupt or unexpected accelerations, or similar hazards.

(g) Acoustical

It shall be resistant to alkalis, greases, gasolines, aliphatic oils, vermin, and dilute acids. The material shall not contain any asbestos, and shall meet the flammability and smoke emission requirements of the TS.

(h) Equipment Enclosures

Fiberglass-reinforced polyester plastic used in underfloor enclosures and doors shall conform to the flammability and smoke emission requirements of the TS.

(i) Electrical Design Standards

Commutating capacitors shall be a paper or plastic film type and incorporate a nontoxic impregnant, and shall be chosen to give a service life of at least 20 years.

(j) 49 CFR 238. 103 : Fire Hazard Analysis will be conducted according to 49 CFR 238.103, and it will be provided.

(k) 49 CFR 238. 103 : Reasonably ensure that a ventilation system in the equipment does not contribute to the lethality of a fire. The sub-contractor will provide the summer (heat) analysis for equipment (device) by the CFD. And sub-contractor will provide the experimental data for equipment (device) summer (heat), if required.

(l) 49 CFR 238. 103 : Identify in writing any train component that is a risk of initiating fire and which requires overheat protection. An overheat detector shall be installed in any component when the analysis determines that an overheat detector is necessary.

(m) Fire Load Analysis

Non-metallic material list will be provided and updated monthly. Non-metallic material list includes all non-metallic material, heat properties (Calorific value), weight, location, etc. The Format of Non-metallic material list will be provided after contract.

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4.3.1 Fire Performance Test Procedure and Criteria

The Fire Performance Test Procedure and Criteria shall be met, but not be limited to, the following Table:

Category	Function of Material	Test Method	Performance Criteria	Remark
Other Vehicle Components	Seat and Mattress Frame, Wall and Ceiling Lining and Panels, Seat and Toilet Shrouds, Trays and Other tables, Partition, Shelves, Opaque Windscreens, End Caps, Roof Housings, and Component Boxes and Cover.	ASTM E 162-98	Is ≤ 35.	See clause 1 – 2 and 9-12 in 49 CFR Part 238 Appendix B.
		ASTM E 662-01	Ds(1.5) ≤ 100, Ds(4.0) ≤ 200.	
		ASTM E 662-01	Ds(1.5) ≤ 100, Ds(4.0) ≤ 175.	
	Thermal & Acoustical Insulation	ASTM E 162-98	Is ≤ 25.	See clause 1 – 2 and 9-12 in

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Category	Function of Material	Test Method	Performance Criteria	Remark
		ASTM E 662-01	Ds(4.0) ≤ 100.	49 CFR Part 238 Appendix B.
Elastomers	Window gaskets, Door nosings, intercar diaphragms, roof mats and seat springs	ASTM C 1166-00	100mm (4.0 in) maximum flame propagation.	See clause 1, 10 and 11 in 49 CFR Part 238 Appendix B.
		ASTM E 662-01	Ds(1.5) ≤ 100, Ds(4.0) ≤ 200.	
Minor material	Individual Small part less than 100 ㎠ (16 in^2)	ASTM E 1354 (Heat Flux : 50 kW/㎡)	q_{180} ≤ 100kW/㎡, $F_{f(Avg)}$ ≤ 500 ㎡/kg.	See clause 11 in 49 CFR Part 238 Appendix B.

The minor materials (Individual Small part less than 100 ㎠ (16 in2)) that are met requirement of the clause 10 and 11 in 49 CFR Part 238 Appendix B.

Should a conflict exist between the requirements of 49 CFR Part 238 and the requirements of 16.2 Smoke and Flammability and requirements listed elsewhere in this Procurement Technical Specification, the more restrictive shall govern.

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4.3.2 Fire Performance Deliverables

The Fire Performance Deliverables shall be provided, but not be limited to, the following Table:

No	TS Section	Description	Due Data	Remark
1	1.21	Fire Load Analysis		
2	1.21, 16.2	Fire Performance Certification		
3	1.21	Combustible material matrix Initial		
4	49 CFR 238. 103	Fire Hazard Analysis		

4.4 RAMS

4.4.1 Reliability Requirement

The Subcontractor shall meet the specified reliability targets under contract considering 61,000 car miles per year for average annual running distance as follows

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Equipment	MDBF (car mile)
Communication System	200,000 miles

Table 1:Reliability Targets

Note: For reliability analysis, average operating speed of 27.5 miles/hour can be used

A chargeable failure shall be defined as any equipment related occurrence considered by DENVER's Railroad Division as rendering the car unfit for service; or as any maintenance action requiring repair or replacement of any subsystem or whole-vehicle component which is not an approved consumable item (or which is approved as being a consumable item but is not achieving its design service life) and which failure has not either been due to a failure occurrence in equipment of another subsystem

4.4.1.1 Fleet Defects (Pattern Failures)

The occurrence of independent failures of the same warranted item that exceeds more than 10 percent of the total number of identical items supplied under the subcontract may be declared a fleet defect or pattern failure. On this basis, the subcontractor shall be required to develop and implement a corrective action program to eliminate the pattern failure. The corrective action program must be approved by Hyundai-Rotem/DENVER and its implementation must be coordinated with Hyundai-Rotem.

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4.4.2 Maintainability Requirement

1) The equipment shall be designed and constructed to have a minimum period of 92 days between periodic maintenance servicing by shop personnel and comply with TS.

2) Subcontract shall propose and demonstrate MTTR (Mean Time To Repair) based on the previous project through maintainability analysis.

4.4.3 Regulation Requirement

Subcontractors shall comply with 49 CFR Part 238.105, APTA's manual, MIL-STD-882C, MIL-STD-1629 and CENELEC standard EN50128, specified in DENVER Hardware and Software Safety Program(Drawing 19-18).

4.4.4 Safety Requirement

1) The Subcontractor shall ensure that the Risk Index of all hazards on the train shall not be worse than the "Acceptable" category i.e. either "G" or B" as defined in the TS of DENVER Hardware and Software Safety Program(Drawing 19-18)

2) The subcontractor shall comply with the general safety design requirement specified in TS.

3) Hazards of Category I or II severity shall be demonstrated by test, the validity of any portion of all analyses.

4.4.5 Failure Analysis

In the event of a failure of any component during contractually required testing or the warranty period, the subcontractor shall submit a complete failure analysis report to Hyundai-Rotem for approval. The failure analysis shall be submitted within 30 days of receipt of the failed part by Subcontractor, unless otherwise agreed upon. Unless otherwise specified, the format of failure report shall be mutually agreed

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4.4.6 RAMS Deliverables

1) Subcontractor shall submit Hardware and Software Safety Program and its supporting analysis to comply with DENVER Hardware and Software Safety Program(Drawing 19-18) for all micro-based safety related systems by within 2 weeks after order by Hyundai-Rotem and it shall include the contents specified in the TS of DENVER Hardware and Software Safety Program(Drawing 19-18).

2) Subcontractor have to submit the Safety Integrity Level Justification Document in order to justify the appropriate SIL rating and for approval

3) The System Hierarchy including LRU(Line Replaceable Unit) and WRU (Workshop Replaceable Unit) list using the attached table by within 2 weeks after order by Hyundai-Rotem

4) The subcontractor shall prepare and submit Reliability and Maintainability Analysis Report at Preliminary Design Review and Final Design Review stage and the contents of it shall consist of, but not be limited to, the following elements

- Definition of Reliability and Maintainability goals and criteria
- Subsystem functional description including subsystem schematics and functional flow diagrams
- A summary of reliability and maintainability prediction results and remedial actions taken as a result of the Reliability and Maintainability Analysis
- A comparison of reliability and maintainability predictions against allocated requirements specified in PTS 2.2.2.1 and 2.2.2.3

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5) Failure Modes and Effects and Criticality Analysis and all electrical circuit failure mode analyses shall include sneak circuit and dry contact analysis at Preliminary Design Review and Final Design Review stage in the TS.

6) Fault Tree Analysis including source data shall be submitted at Preliminary Design Review and Final Design Review stage in the TS.

The Hazards of Category I or II severity shall be subject to Fault Tree Analysis.

7) RAM data table using attached file shall be submitted at Preliminary Design Review and Final Design Review stage in the TS.

8) The subcontractor shall prepare and submit Safety Analysis Report at Preliminary Design Review and Final Design Review stage in the TS and the contents of it shall consist of, but not be limited to, the following elements

- Definition of safety goals and criteria
- Subsystem functional description including subsystem schematics and functional flow

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diagrams

- Description of Safety Feature
- A summary of the objectives, findings and remedial actions taken as a result of the Safety Analysis
- Hazard Analysis including Preliminary Hazard Analysis(PHA), Sub-system Hazard Analysis, System Hazard Analysis.

4.5 Software

Subcontractor shall comply with the following DENVER's requirement.

- Clause XXXV and XXXVI of DENVER's RFP
- Clause TS of DENVER.
- DENVER Hardware and Software Safety Program: Subcontactor shall submit Hardware and Software Safety Program
- Subcontractor shall submit following software documents as CDRL submissions.
- The below CDRL no and title will be decided.

For single documentation methodologies, Subcontractor shall follow the contents of each software documents guided in international standard as follows

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Title	Contents	Remark
Software Quality Assurance Plan	Clause 4 of IEEE-STD-730	
System Functional Description		
Software Requirement Specification	Clause 5 of IEEE-STD-830	
Software Design Description	Annex A of IEEE-STD-1016 ATA No.102	
Software Verification and Validation Plan	Clause 7 of IEEE-STD-1012	
Software Verification and Validation Report	Clause 7.6 of IEEE-STD-1012	
Software Configuration Management Plan	Clause 4 of IEEE-STD-828	
Software User Documentation	Clause 5 of Software User Documentation	

All software supplied for the system designed structurally and logically, fully documented, thoroughly tested in a systematic manner such that they can achieve high quality in terms of safety, reliability, testability, traceability and maintainability. Also all software supplied for the system shall be developed in line with latest ISO 9000-3.

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Subcontractor have to submit the SIL Justification Document in order to justify the appropriate SIL rating and for approval.

For the adoption of EN50128 with SIL 1 and above, ISA shall perform software assessment in compliance with clause 14 of EN50128. Finally, Subcontractor shall submit final Software Assessment Report so that Hyundai-Rotem can ensure that assigned SIL has been achieved.

Subcontractor has to comply with Hyundai-Rotem's Software Quality Assurance Plan which will be approved by DENVER

Especially, clause 16.14.7(Escrow Account), the Escrow Agreement should be made among depositor(Subcontractor), Beneficiary(DENVER) and Escrow Agent. Subcontractor shall have all responsible for making arrangement for Escrow Agreement including all fees for all Escrow Services, discussion with DENVER and etc. Subcontractor must comply with DENVER's proposed standard escrow agreement document. The deposit escrow period is 15 years from software stabilization(e.g. warranty period)

Subcontractor shall provide the properties and all requested material for Hyundai-Rotem's software quality & safety audit to Subcontractor. Corrective action against NCR and OBS should be returned to Hyundai-Rotem within one week after software audit. And also, Implementation of corrective actions against to Identified NCR(Non conformance Report) and OBS(Observations) should be finalized within one-month after software audit. Hyundai-Rotem will not pay the amount of money for software development without the completion of the identified NCR and OBS.

The Subcontractor shall be obliged to take care of any software change if Hyundai-Rotem and DENVER request the change for the correction of software during commissioning phase, commissioning and Warranty Period, etc. The change shall be implemented by the Subcontractor. In the event, when any non-conformity arises to the specified requirements, the Subcontractor shall take remedial measures at its own cost within the schedule to be agreed with Hyundai-Rotem.

The Subcontractor shall be obliged to abide by the software change control procedure on site(or online based change control system(http://sccp.Hyundai-Rotem.co.kr) which should be prepared by Hyundai-Rotem in order to enhance the regulation of software change in train-set. It should be not allowed to upload the changed software into train-set without the approval by Hyundai-Rotem/ DENVER under software change control by Hyundai-Rotem. Subcontractor has to supply test report of system integration testing or bench testing of the updated software performed subcontractor's premises before formal release of the updated software.

The Subcontractor shall provide Diagnostic & Maintenance Software and it's user manual.

4.6 Noise

Not Applicable

5. Design Information

In order to take an approval for the communication system from DENVER, the subcontractor shall provide all technical information as a written document. The below document/deliverable shall be provided as a minimum.

All technical information shall follow the "publications style guide" attached in TS.

5.1 General

The Subcontractor shall submit, the following general information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline

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1	Test and Inspection plan		At DDP
2	As-built drawing		At DDP
3	Operation & Maintenance Manual		At DDP
4	Illustrated parts catalog		At DDP
5	Training plan		At DDP
6	Training material		At DDP
7	List of warranty period spare parts		At DDP
8	List of recommended spare part		At DDP
9	List of special tools and test equipment		At DDP
10	PCB information		At DDP
11	Software Source code		At DDP

* Note: The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem.

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5.2 Design

The Subcontractor shall submit, the following design information in accordance with the schedule:

No	Document / Deliverables	Reference	*Deadline
1	Clause by Clause Compliance sheet for the Technical Specification of DENVER		Before Contraction
2	Dimensional Assembly Drawing by Autocad file.		At CDP
3	Sub-Assembly Drawing		At CDP
4	System description		At CDP
5	Function description		At CDP
6	Product description		At CDP
7	System Block Diagram or Schematic Diagram		At CDP
8	Installation Specification of On-board equipment		At CDP
9	Installation Specification and investigation report of Wayside equipment		At CDP
10	Interface Diagram and Specification with CDT		At CDP
11	Interface Diagram and Specification with Wayside Equipment		At CDP
12	Cabling requirement		At CDP
13	Test Specification		At DDP

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14	Test Report		At DDP
15	Test Specification of Communication System at Complete Vehicle		At DDP
16	Test Report of Complete Vehicle		At DDP
17	Commissioning Test Procedure for Communication System (Including interface test with Wayside Equipment)		At DDP
18	Test Report of Commissioning Test		At DDP

* Note:

1. The deadline to be discussed and agreed between the Subcontractor and Hyundai-Rotem

5.3 SEM

The subcontractor shall provide all design information requested in the clause 4 of this PTS for EMI/EMC, Fire, RAMS and Software.

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6. Tests and Inspection

The subcontractor shall perform all tests specified in the TS. The test procedure and the report shall be prepared and provided by the subcontractor.

7. Attachment

DENVER RTD EMU_ TS

CONTRACT PART1.pdf

Attachment 7 .pdf

DENVER RTD SEMS REQUIREMENT

Attachment_SEMS Requirement.zip

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